

3D-printed acoustic holograms move objects in midair

A team from Max Planck Institute for Intelligent Systems, Germany, has described a new way of easily creating very high-resolution acoustic holograms that work in air or water. Rather than relying on a whole bunch of small transducers, these use just one giant transducer that sits underneath a special 3D-printed transmission hologram made out



3D-printed plastic blocks can now be turned into acoustic holograms that generate 3D shapes made of sound, which could function like sonic tractor beams (Image courtesy: <https://1news.in>)

of finely-contoured plastic. The single transducer emits one type of sound wave, which means that you cannot use it to create a sound field that will do all that much for you.

The trick is to use a 3D-printed piece of plastic. The block is attached to the transducer, and when it transduces, the sound wave has to propagate through the block before it is emitted into space. Since sound moves slower through the block than it does through air or water, by carefully printing the top of the block in a very specific pattern to vary its thickness, you get the same effect as using a transducer array made up of individual transducers that are each as small as the resolution of your 3D printer. This allows creation of acoustic holograms that are about 100 times as detailed as anyone has been able to make before.

Advanced imaging technique for cancer detection

Researchers from University of Cambridge are developing a new imaging technique with the aim of detecting and characterising early cancerous changes in the gastrointestinal tract. The technique involves using a standard endoscopy system with a novel set of camera filters, increasing the number of colours that can be visualised during endoscopy, and potentially improving the ability to detect abnormal cells in the lining of the gut.

Dr Sarah Bohndiek from University of Cambridge, UK, who is leading the development, explains the new technique as, "In traditional endoscopy, we use white light and detectors that replicate our eyes, which detect light in red, green and blue colour channels. We are now developing an approach called hyperspectral imaging, which will increase the number of colour channels that can be visualised from three to over 50."

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